CPSC 233 Assignments 02 **Due Date: July 16th by 11:59 PM**

Topic: Introduction to Classes (creating classes, copy constructors)

Collaboration

Discussing the assignment requirements with others is a reasonable thing to do, and an excellent way to learn. However, the work you hand-in must ultimately **be your work**. This is essential for you to benefit from the learning experience, and for the instructors and TAs to grade you fairly. Handing in work that is not your original work, but is represented as such, is plagiarism and academic misconduct. Penalties for academic misconduct are outlined in the university calendar.

Here are some tips to avoid plagiarism in your programming assignments.

1. Cite all sources of code that you hand-in that are not your original work. You can put the citation into comments in your program. For example, if you find and use code found on a web site, include a comment that says, for example:
   `# the following code is from https://www.quackit.com/python/tutorial/python_hello_world.cfm.`
   Use the complete URL so that the marker can check the source.
2. Citing sources avoids accusations of plagiarism and penalties for academic misconduct. However, you may still get a low grade if you submit code that is not primarily developed by yourself.
3. Discuss and share ideas with other programmers as much as you like, but make sure that when you write your code that it is your own. A good rule of thumb is to wait 20 minutes after talking with somebody before writing your code. If you find yourself exchanging code by electronic means, writing code while sitting and discussing with a fellow student, typing what you see on another person's console, then you can be sure that "your" code is not substantially your own, and your sources must then be cited to avoid plagiarism.
4. **Collaborative coding is strictly prohibited.** Your assignment submission must be strictly your code. Discussing anything beyond assignment requirements and ideas is a strictly forbidden form of collaboration. This includes sharing code, discussing code itself, or modeling code after another student's algorithm. **You cannot use (even with citation) another student's code.**
5. We will be looking for plagiarism in all code submissions, possibly using automated software designed for the task. For example, see Measures of Software Similarity (MOSS - https://theory.stanford.edu/~aiken/moss/).

Remember, if you are having trouble with an assignment, it is always better to go to your TA and/or instructor to get help, than it is to plagiarize.

It is important to note that your code should be able to run from the command line. Your TA should not install/delete any packages/module to run the program.

If you are facing any technical difficulty especially using Lab machines, send your email to Shokoofa zahra.ghods@ucalgary.ca. She is also added in the Discord server.

Requirements:

For this assignment, you will create one class: Vehicle (use the attached starter code). Make sure to use the name of the class as I have explicitly mentioned.

Below are the detailed requirements for the Vehicle class

Vehicle:

It will have three instance variables: make, model and mileage. Below are the details of the three instance variables

- make represents the maker of the vehicle such as Toyota, Honda
- model represents the model of the vehicle such as Corolla, Camry, Civic
- mileage should be non-negative that has value between 0 and 25 (both are inclusive)

Your vehicle class should have two constructors:

- A constructor that initializes the instance variables at the time of the object creation
- A copy constructor

Your vehicle class should have the following method:

- ***getFuelEfficiencyCategory:*** Takes no argument and returns a String representing the fuel efficiency category which is based on their mileage. A vehicle is considered *best* category if its mileage is greater than 15 (exclusive), *average* category if the mileage is between 15 and 8 (both values are inclusive), *worst* category otherwise.

In addition to above methods, you need to have getters and setters methods for the instance variable too. You will write three setters methods (setMake, setModel, setMileage) that sets each instance variable, and three get (getMake, getModel, getMileage) methods, that return each instance variables.

A template for the vehicle class is provided. You should write code in that file. There should be no change in the main function. Also make sure that your class names, methods names are correct.

Submission:

Use the University of Calgary Desire2Learn system (http://d2l.ucalgary.ca) to submit your assignment work online. Log in using your UofC eID and password, then find our course, cpsc 233, in the list. Then navigate to *Assessments -> Dropbox Folders*, and find the folder for Assignment #2 here. You should submit the following files

1. Vehicle class code file